TE3 Worked Examples: Solving Trigonometric Equations

Recording companion for the TE3 CheckIt support video — Math 172, Santa Ana College

How to use this document

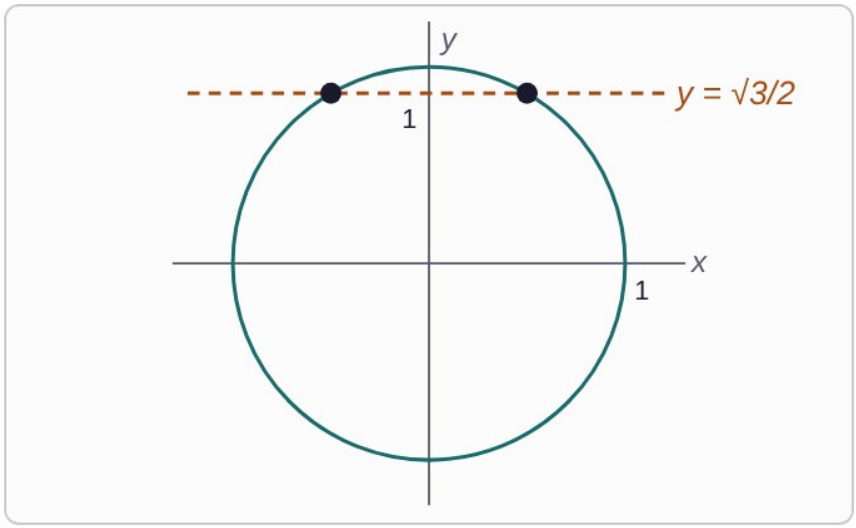
- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Where on the unit circle does this function take this value? Isolate the trig function, then find every point on the unit circle where it has that value. Sine is the y-coordinate, cosine is the x-coordinate, and tangent is y over x. Most values happen at two angles in [0, 2π). For all solutions, add 2πk to each one. A square gives plus or minus, so up to four angles. And for cos(3θ), let the angle 3θ go around three times before dividing by 3.

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Example 1: Unit Circle 1



Plotted data: The unit circle, centered at the origin with radius 1. A dashed horizontal line y = square root of 3 over 2 crosses the circle at two points, one on each side of the y-axis, both above the x-axis. The two points are marked with dots but not labeled.

$$2\sin\theta-\sqrt{3}=0$$

Question

Which angles in [0, 2π) solve this equation?

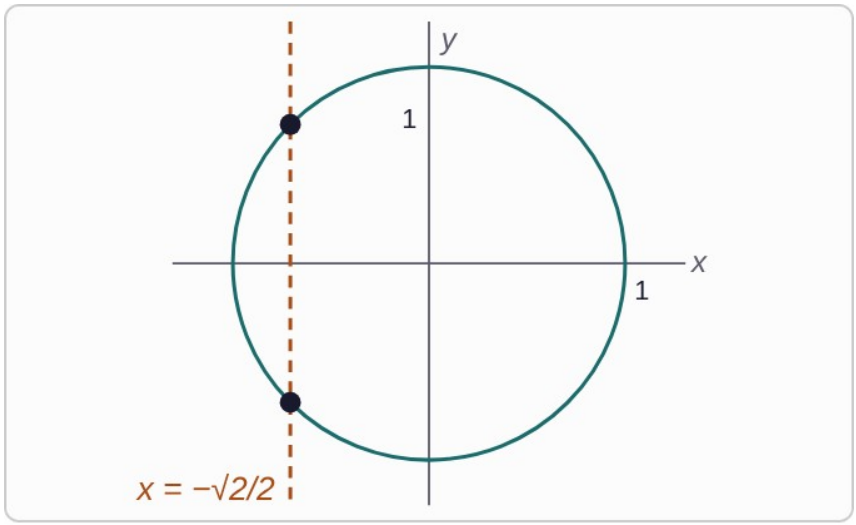
- **A.** $\pi/6$, $(5\pi)/6$
- **B.** $\pi/3$, $(2\pi)/3$
- **C.** $\pi/3$, $(5\pi)/3$
- **D.** $\pi/3$

Answer and reasoning

Isolate: $\sin\theta=(\sqrt{3})/2$. Sine is the y-coordinate, and the line $y=(\sqrt{3})/2$ crosses the circle twice: at $\pi/3$ in Quadrant I and at $\pi-\pi/3=(2\pi)/3$ in Quadrant II. The pair $\pi/6, (5\pi)/6$ has sine $1/2$, and $(5\pi)/3$ has a negative sine. Stopping at $\pi/3$ misses the second crossing. ***θ=π/3, (2π)/3*.**

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Example 2: Unit Circle 2



Plotted data: The unit circle, centered at the origin with radius 1. A dashed vertical line x = negative square root of 2 over 2 crosses the circle at two points, both to the left of the y-axis, one above and one below the x-axis. The two points are marked with dots but not labeled.

$$2\cos\theta+\sqrt{2}=0$$

Question

Which angles in [0, 2π) solve this equation?

- **A.** $(2\pi)/3$, $(4\pi)/3$
- **B.** $(3\pi)/4$, $(5\pi)/4$
- **C.** $(5\pi)/6$, $(7\pi)/6$
- **D.** $(3\pi)/4$

Answer and reasoning

Isolate: $\cos\theta=-(\sqrt{2})/2$. Cosine is the x-coordinate, so the vertical line $x=-(\sqrt{2})/2$ crosses the circle on the left side, once above and once below the x-axis: $(3\pi)/4$ and $(5\pi)/4$. The pairs $(2\pi)/3$, $(4\pi)/3$ and $(5\pi)/6$, $(7\pi)/6$ are on the left side too, but their cosines are $-1/2$ and $-(\sqrt{3})/2$. Stopping at $(3\pi)/4$ misses the crossing below the axis. ***θ=(3π)/4, (5π)/4*.**

Example 3: Unit Circle 3

$$3\sec\theta - 2\sqrt{3} = 0$$

Question

Which angles in [0, 2π) solve this equation?

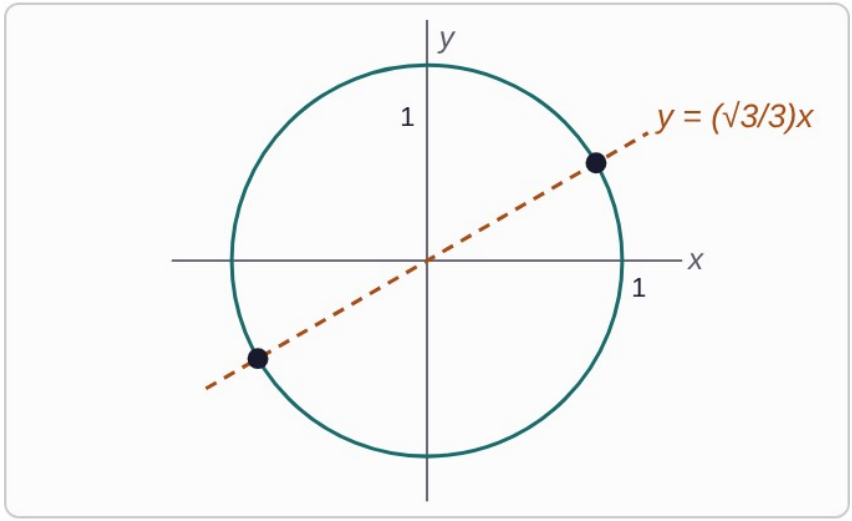
- **A.** $(7\pi)/6$, $(11\pi)/6$
- **B.** No solution: cosine cannot be more than 1
- **C.** $\pi/6$, $(11\pi)/6$
- **D.** $\pi/3$, $(5\pi)/3$

Answer and reasoning

Isolate: $\sec\theta = (2\sqrt{3})/3$. Flip to cosine and rationalize: $\cos\theta = 3/(2\sqrt{3}) = (\sqrt{3})/2$. That happens at $\pi/6$ and at $2\pi - \pi/6 = (11\pi)/6$. It is secant that equals $(2\sqrt{3})/3 \approx 1.15$; after flipping, the cosine is about 0.87, which is fine. The pair $(7\pi)/6$, $(11\pi)/6$ has sine $-1/2$, and $\pi/3$, $(5\pi)/3$ has cosine $1/2$. ***θ = π/6, (11π)/6*.**

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Example 4: Unit Circle 4



Plotted data: The unit circle, centered at the origin with radius 1. A dashed line through the origin, y = square root of 3 over 3 times x, rises gently to the right. It crosses the circle at two opposite points, one up and to the right and one down and to the left. The two points are marked with dots but not labeled.

$$3\tan\theta-\sqrt{3}=0$$

Question

Which angles in [0, 2π) solve this equation?

- **A.** $\pi/6$, $(7\pi)/6$
- **B.** $\pi/3$, $(4\pi)/3$
- **C.** $\pi/6$, $(5\pi)/6$
- **D.** $\pi/6$, $(11\pi)/6$

Answer and reasoning

Isolate: $\tan\theta=(\sqrt{3})/3$. Tangent is y/x, the slope of the line from the origin, so the line $y=((\sqrt{3})/3)x$ crosses the circle at two opposite points: $\pi/6$ and $\pi/6+\pi=(7\pi)/6$. Tangent angles always come in pairs π apart. The pair $\pi/3$, $(4\pi)/3$ has tangent $\sqrt{3}$, and the other two pairs use sine or cosine symmetry. ***θ=π/6, (7π)/6*.**

Example 5: All Solutions 1

$$2\sin\theta - 1 = 0$$

Question

Which describes all solutions? Here k is any integer.

A. $\pi/6 + 2\pi k$, $(5\pi)/6 + 2\pi k$
B. $\pi/6 + 2\pi k$
C. $\pi/6 + \pi k$
D. $\pi/6 + 2\pi k$, $(11\pi)/6 + 2\pi k$

Answer and reasoning

$\sin\theta = 1/2$ at $\pi/6$ and $(5\pi)/6$. Going around the circle any whole number of times lands on the same points, so add $2\pi k$ to each. Keeping only $\pi/6$ loses half the solutions. The list $\pi/6 + \pi k$ includes $(7\pi)/6$, where sine is $-1/2$. The angle $(11\pi)/6$ has a negative sine. ***$\theta = \pi/6 + 2\pi k,\ (5\pi)/6 + 2\pi k$*.**

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Example 6: All Solutions 2

$$\tan\theta+1=0$$

Question

Which describes all solutions? Here k is any integer.

A. $(3\pi)/4+2\pi\ k$
B. $\pi/4+2\pi\ k$, $(5\pi)/4+2\pi\ k$
C. $(3\pi)/4+2\pi\ k$, $(5\pi)/4+2\pi\ k$
D. $(3\pi)/4+2\pi\ k$, $(7\pi)/4+2\pi\ k$

Answer and reasoning

$\tan\theta=-1$ where x and y are opposites: $(3\pi)/4$ and $(7\pi)/4$, which are π apart. So all solutions are $(3\pi)/4+2\pi\ k$, $(7\pi)/4+2\pi\ k$, or more briefly $(3\pi)/4+\pi\ k$. CheckIt writes the longer form. Keeping only $(3\pi)/4+2\pi\ k$ misses $(7\pi)/4$. The pair $\pi/4$, $(5\pi)/4$ has tangent $+1$, and so does $(5\pi)/4$ in the remaining choice. $\boldsymbol{\theta=(3\pi)/4+2\pi\ k,\ (7\pi)/4+2\pi\ k}$**.**

Example 7: All Solutions 3

$$3\cot\theta-\sqrt{3}=0$$

Question

Which describes all solutions? Here k is any integer.

A. $\pi/3+2\pi k$
B. $\pi/6+2\pi k$, $(7\pi)/6+2\pi k$
C. $\pi/3+2\pi k$, $(2\pi)/3+2\pi k$
D. $\pi/3+2\pi k$, $(4\pi)/3+2\pi k$

Answer and reasoning

Isolate: $\cot\theta=(\sqrt{3})/3$. Flip: $\tan\theta=3/(\sqrt{3})=\sqrt{3}$, at $\pi/3$ and $\pi/3+\pi=(4\pi)/3$. The pair $\pi/6$, $(7\pi)/6$ solves $\tan\theta=(\sqrt{3})/3$, forgetting to flip. At $(2\pi)/3$ the tangent is $-\sqrt{3}$, and keeping only $\pi/3+2\pi k$ loses half the solutions. ***θ=π/3+2π k, (4π)/3+2π k*.**

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Example 8: All Solutions 4

$$\csc\theta+2=0$$

Question

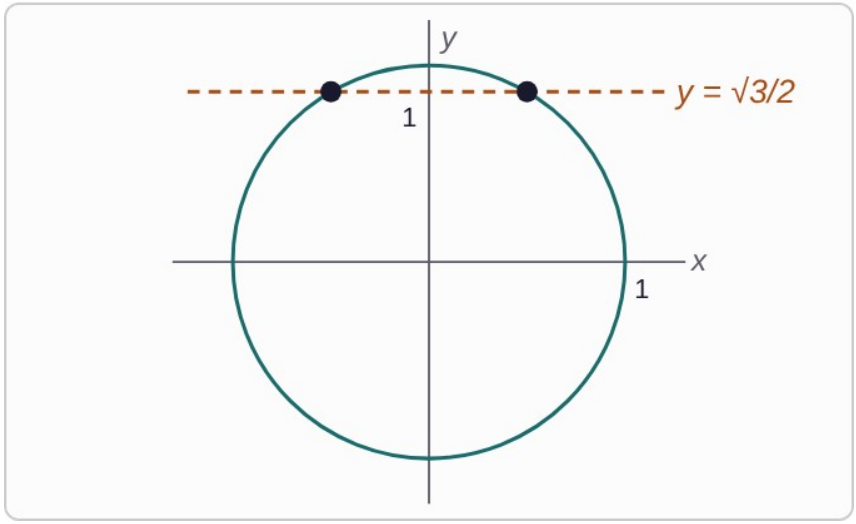
Which describes all solutions? Here k is any integer.

A. $\pi/6+2\pi\, k$, $(5\pi)/6+2\pi\, k$
B. $(7\pi)/6+2\pi\, k$, $(11\pi)/6+2\pi\, k$
C. $(4\pi)/3+2\pi\, k$, $(11\pi)/6+2\pi\, k$
D. No solution: cosecant cannot be negative

Answer and reasoning

$\csc\theta=-2$ means $\sin\theta=-1/2$. Cosecant can be negative; it just can never be strictly between −1 and 1. Sine is $-1/2$ below the x-axis, at $\pi+\pi/6=(7\pi)/6$ and $2\pi-\pi/6=(11\pi)/6$. The pair $\pi/6,\ (5\pi)/6$ has sine $+1/2$, and at $(4\pi)/3$ the sine is $-(\sqrt{3})/2$. **$\theta=(7\pi)/6+2\pi\, k,\ (11\pi)/6+2\pi\, k$.**

Example 9: Squared 1



Plotted data: The unit circle, centered at the origin with radius 1. A dashed horizontal line y = square root of 3 over 2 crosses the circle at two points above the x-axis, one on each side of the y-axis. The points are marked with dots but not labeled.

$$4\sin^2\theta - 3 = 0$$

Question

Which angles in [0, 2π) solve this equation?

- **A.** $\pi/3$, $(2\pi)/3$
- **B.** $\pi/6$, $(5\pi)/6$, $(7\pi)/6$, $(11\pi)/6$
- **C.** $\pi/3$, $(2\pi)/3$, $(4\pi)/3$, $(5\pi)/3$
- **D.** $\pi/3$, $(4\pi)/3$

Answer and reasoning

$\sin^2\theta = 3/4$, so $\sin\theta = \pm\sqrt{3}/2$. The plus gives $\pi/3$, $(2\pi)/3$, the two points in the picture, and the minus gives $(4\pi)/3$, $(5\pi)/3$. Keeping only the plus loses half. The list in sixths has cosine $\pm\sqrt{3}/2$, and $\pi/3$, $(4\pi)/3$ is a tangent pair. ***$\theta = \pi/3, (2\pi)/3, (4\pi)/3, (5\pi)/3$*.**

Example 10: Squared 2

$$2\cos^2\theta - 1 = 0$$

Question

Which angles in [0, 2π) solve this equation?

- **A.** $\pi/4$, $(3\pi)/4$, $(5\pi)/4$, $(7\pi)/4$
- **B.** $\pi/4$, $(7\pi)/4$
- **C.** $\pi/3$, $(2\pi)/3$, $(4\pi)/3$, $(5\pi)/3$
- **D.** $\pi/4$, $(5\pi)/4$

Answer and reasoning

$\cos^2\theta = 1/2$, so $\cos\theta = \pm\sqrt{(1/2)} = \pm(\sqrt{2})/2$. Cosine is $\pm(\sqrt{2})/2$ at all four angles in fourths that are not on an axis. The pair $\pi/4, (7\pi)/4$ keeps only the plus. The list in thirds forgets the square root, solving $\cos\theta = \pm 1/2$, and $\pi/4, (5\pi)/4$ is a tangent pair. ***θ=π/4, (3π)/4, (5π)/4, (7π)/4*.**

Example 11: Squared 3

$$\cos^2\theta - 1 = 0$$

Question

Which angles in [0, 2π) solve this equation?

- **A.** 0
- **B.** $\pi/2$, $(3\pi)/2$
- **C.** 0, π, 2π
- **D.** 0, π

Answer and reasoning

$\cos^2\theta = 1$, so $\cos\theta = \pm 1$: $\theta = 0$ gives 1 and $\theta = \pi$ gives −1. Here the plus or minus gives two angles, not four, because each value happens only once on the circle. 2π is not in $[0, 2\pi)$, since it is the same point as 0. The pair $\pi/2$, $(3\pi)/2$ is where cosine is 0. ***θ=0, π.***

Example 12: Squared 4

$$4\sin^2\theta - 1 = 0$$

Question

Which angles in [0, 2π) solve this equation?

- **A.** $\pi/6$, $(5\pi)/6$
- **B.** $\pi/6$, $(5\pi)/6$, $(7\pi)/6$, $(11\pi)/6$
- **C.** $\pi/3$, $(2\pi)/3$, $(4\pi)/3$, $(5\pi)/3$
- **D.** $\pi/6$, $(5\pi)/6$, $(11\pi)/6$, $(13\pi)/6$

Answer and reasoning

$\sin^2\theta = 1/4$, so $\sin\theta = \pm 1/2$: $\pi/6$, $(5\pi)/6$ for the plus and $(7\pi)/6$, $(11\pi)/6$ for the minus. The list in thirds has cosine $\pm 1/2$, and $(13\pi)/6$ is past 2π (it is the same point as $\pi/6$), while $(7\pi)/6$ is missing. ***θ = π/6, (5π)/6, (7π)/6, (11π)/6*.**

Example 13: Multiple Angle 1



Plotted data: Two circles side by side. Each arrow starts on the right side of its circle and turns counterclockwise. On the left, an arrow goes once around a circle, labeled theta from 0 to 2 pi, once around. On the right, an arrow spirals twice around a circle, labeled 2 theta from 0 to 4 pi, twice around.

$$2\cos(2\theta)-\sqrt{3}=0$$

Question

Which angles in [0, 2π) solve this equation?

- **A.** $\pi/12$, $(11\pi)/12$
- **B.** $\pi/6$, $(11\pi)/6$
- **C.** $\pi/12$, $(11\pi)/12$, $(13\pi)/12$, $(23\pi)/12$
- **D.** $\pi/12$, $(11\pi)/12$, $(13\pi)/12$, $(23\pi)/12$, $(25\pi)/12$

Answer and reasoning

$\cos(2\theta)=(\sqrt{3})/2$. As θ goes around once, 2θ goes from 0 to 4π, twice around. So list the angles for 2θ on two laps: $\pi/6, (11\pi)/6, (13\pi)/6, (23\pi)/6$. Then divide each by 2. Stopping after one lap gives only two; not dividing gives $\pi/6, (11\pi)/6$; and $(25\pi)/12$ is more than 2π. ***θ=π/12, (11π)/12, (13π)/12, (23π)/12*.**

Example 14: Multiple Angle 2

$$2\sin(3\theta)-\sqrt{2}=0$$

Question

Which angles in [0, 2π) solve this equation?

- **A.** $\pi/12$, $\pi/4$, $(3\pi)/4$, $(11\pi)/12$, $(17\pi)/12$, $(19\pi)/12$, $(25\pi)/12$
- **B.** $\pi/4$, $(3\pi)/4$
- **C.** $\pi/12$, $\pi/4$, $(3\pi)/4$, $(11\pi)/12$
- **D.** $\pi/12$, $\pi/4$, $(3\pi)/4$, $(11\pi)/12$, $(17\pi)/12$, $(19\pi)/12$

Answer and reasoning

$\sin(3\theta)=(\sqrt{2})/2$. Now 3θ goes from 0 to 6π, three laps. Each lap has two angles: $\pi/4$, $(3\pi)/4$, then $(9\pi)/4$, $(11\pi)/4$, then $(17\pi)/4$, $(19\pi)/4$. Divide each by 3. Two laps give only four answers, and $(25\pi)/12$ is past 2π, from a fourth lap. Not dividing gives $\pi/4$, $(3\pi)/4$. ***θ=π/12, π/4, (3π)/4, (11π)/12, (17π)/12, (19π)/12*.**

Example 15: Multiple Angle 3

$$\sin(2\theta)=0$$

Question

Which angles in [0, 2π) solve this equation?

- **A.** $0, \pi/2$
- **B.** $\pi/2, \pi, (3\pi)/2$
- **C.** $0, \pi/2, \pi, (3\pi)/2$
- **D.** $0, \pi/2, \pi, (3\pi)/2, 2\pi$

Answer and reasoning

Sine is 0 at 0 and π, once per half lap. With two laps, $2\theta=0, \pi, 2\pi, 3\pi$, and dividing by 2 gives $0, \pi/2, \pi, (3\pi)/2$. Remember that 0 counts: it is in $[0,2\pi)$. One lap gives only $0,\pi/2$, and 2π is not in the interval. ***θ=0, π/2, π, (3π)/2*.**

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Example 16: Multiple Angle 4

$$2\sin(2\theta)-\sqrt{3}=0$$

Question

Which angles in [0, 2π) solve this equation?

- **A.** $\pi/6$, $\pi/3$, $(7\pi)/6$, $(4\pi)/3$
- **B.** $\pi/6$, $\pi/3$
- **C.** $\pi/3$, $(2\pi)/3$
- **D.** $\pi/3$, $(2\pi)/3$, $(7\pi)/3$, $(8\pi)/3$

Answer and reasoning

$\sin(2\theta)=\sqrt{3}/2$, and 2θ makes two laps: $\pi/3, (2\pi)/3, (7\pi)/3, (8\pi)/3$. Divide by 2: $\pi/6, \pi/3, (7\pi)/6, (4\pi)/3$. The list $\pi/3, (2\pi)/3, (7\pi)/3, (8\pi)/3$ is the right list for 2θ, before dividing. The pair $\pi/6, \pi/3$ stops after one lap, and $\pi/3, (2\pi)/3$ is one lap without dividing. ***$\theta=\pi/6, \pi/3, (7\pi)/6, (4\pi)/3$*.**

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Example 17: Slip 1

$$\cos^2\theta = 3/4$$
$$\cos\theta = (\sqrt{3})/2$$
$$\theta = \pi/6,\ (11\pi)/6$$

Question

A student solved $4\cos^2\theta - 3 = 0$ on $[0, 2\pi)$ with this work. What is the mistake, if any?

A. There is no mistake.
B. It should be $\sin\theta = \sqrt{3}/2$ instead, giving $\pi/3$ and $2\pi/3$ for θ.
C. The square root gives $\pm\sqrt{3}/2$, so $5\pi/6$ and $7\pi/6$ are missing.
D. It should be $\cos\theta = 3/4$, with no square root taken at all here.

Answer and reasoning

Taking the square root gives $\cos\theta = \pm(\sqrt{3})/2$. The minus adds $(5\pi)/6$ and $(7\pi)/6$, on the left side of the circle. ***$\theta = \pi/6,\ (5\pi)/6,\ (7\pi)/6,\ (11\pi)/6$*.**

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Example 18: Slip 2

$$\tan\theta=\sqrt{3}$$
$$\theta=\pi/3+\pi\, k$$

Question

A student found all solutions of $\tan\theta-\sqrt{3}=0$ this way, for all integers k. What is the mistake, if any?

A. There is no mistake.
B. It should be π/3 + 2πk, since the period is 2π.
C. It should also include 2π/3 + πk.
D. The angle should be π/6, not π/3.

Answer and reasoning

$\tan(\pi/3)=(\sqrt{3}/2)/(1/2)=\sqrt{3}$, and tangent repeats every π, so $\pi/3+\pi\, k$ covers both $\pi/3$ and $(4\pi)/3$. CheckIt would write $\pi/3+2\pi\, k$, $(4\pi)/3+2\pi\, k$, which is the same set. **There is no mistake.**

Example 19: Slip 3

$$\sin(2\theta)=1/2$$
$$2\theta=\pi/6,\ (5\pi)/6$$
$$\theta=\pi/12,\ (5\pi)/12$$

Question

A student solved $2\sin(2\theta)-1=0$ on $[0, 2\pi)$ with this work. What is the mistake, if any?

A. It should multiply by 2 instead, giving $\pi/3$ and $5\pi/3$ for θ.
B. There is no mistake.
C. The second answer should be $7\pi/12$, not $5\pi/12$, for θ.
D. 2θ goes around twice, so $13\pi/12$ and $17\pi/12$ are missing.

Answer and reasoning

For θ in $[0,2\pi)$, 2θ is in $[0,4\pi)$, so the second lap adds $2\theta=(13\pi)/6,\ (17\pi)/6$. Dividing by 2 gives two more answers. ***$\theta=\pi/12,\ (5\pi)/12,\ (13\pi)/12,\ (17\pi)/12$*.**

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Example 20: Slip 4

$$\sec\theta = 1/2$$
$$\theta = \pi/3,\ (5\pi)/3$$

Question

A student solved $2\sec\theta - 1 = 0$ on $[0, 2\pi)$ with this work. What is the mistake, if any?

A. It should be $2\pi/3$ and $4\pi/3$, where cosine is $-1/2$, instead.
B. $\sec\theta = 1/2$ means $\cos\theta = 2$, so there is no solution.
C. There is no mistake.
D. It should be $\pi/6$ and $11\pi/6$, from $\cos\theta = \sqrt{3}/2$, instead.

Answer and reasoning

Secant is the reciprocal of cosine: $\sec\theta = 1/2$ means $\cos\theta = 2$. Cosine is always between -1 and 1, so no angle works. The student used $\cos\theta = 1/2$, which is a different equation. **There is no solution.**

Example 21: Slip 5

$$\cos\theta=(\sqrt{3})/2$$
$$\theta=\pi/6+2\pi\, k$$

Question

A student found all solutions of $2\cos\theta-\sqrt{3}=0$ this way, for all integers k. What is the mistake, if any?

A. There is no mistake.
B. It is missing the second angle, 11π/6 + 2πk.
C. It should be π/6 + πk, since cosine repeats every π.
D. The angle should be π/3, not π/6.

Answer and reasoning

Cosine is $(\sqrt{3})/2$ at two points: $\pi/6$ and $2\pi-\pi/6=(11\pi)/6$. Each needs its own $2\pi\, k$. Cosine repeats every 2π, not π, so $\pi/6+\pi\, k$ would include $(7\pi)/6$, where cosine is negative. ***θ=π/6+2π k, (11π)/6+2π k*.**